

The Lunar Heat-Flow Study: A Guidebook from the Ground Up

A plain-language companion to
“Difference of Lunar Regolith Thermal Conductivity K_d at the Apollo 15 and 17 Heat-Flow Boreholes

June 15, 2026

Who this book is for

You. Even if you have never taken a physics or statistics course. Every chapter starts from zero and builds up. The coloured boxes are your friends: **plain-words** boxes restate the idea simply, **equation** boxes unpack every symbol of a formula, and **key-takeaway** boxes summarise what to remember. You can read the whole thing skipping every equation box and still understand the study; the boxes are there for when you want to go deeper.

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Part I

Foundations

Chapter 1

Why anyone studies heat in the Moon

The Moon looks dead and unchanging, but just under its surface there is a slow, quiet flow of heat — and that flow carries information we can get no other way.

The surface vs. the inside. The Moon’s surface is baked by the Sun during its two-week-long day and freezes during its two-week night. But a metre below the surface, none of that daily drama is felt. Down there, the temperature is set by two things only: how well the soil conducts heat, and a faint warmth seeping up from the Moon’s deep interior. Measuring that quiet underground temperature tells us about both.

The number we are chasing. The single most important property of the soil, for this purpose, is its **thermal conductivity**: how easily heat passes through it. We write it K , and the specific value deep down (where the soil is packed and settled) is K_d (“ d ” for deep). Lunar soil is an extraordinary insulator — several times better than the styrofoam in a coffee cup — which is exactly why the daily heat wave cannot get far below the surface.

Why it matters in practice. Today, scientists who model the Moon use *one single value of K_d for the entire globe*. That value was never checked against direct underground measurements, because such measurements exist at only two spots in all of history: the Apollo 15 and Apollo 17 landing sites, where astronauts buried thermometers in 1971–72. Every spacecraft that infers ground properties from orbit, and every future mission that wants to “see” below the surface, leans on that one assumed number.

Key takeaway

This study uses the only two underground temperature records ever taken on the Moon to test a hidden assumption everyone relies on: *is one conductivity value good enough for the whole Moon?* The short answer we will build toward: no — the two sites differ by roughly a factor of two, and that difference is real.

Chapter 2

How heat moves: conduction from scratch

2.1 Heat flows downhill (in temperature)

Heat is just jiggling atoms. Where atoms jiggle a lot, it is hot; where they jiggle little, it is cold. When a hot region touches a cold region, the jiggling spreads out: heat always flows from hot to cold. This spreading-through-solid-material is called **conduction** (Figure 2.1).

2.2 Fourier’s law: the rate of heat flow

How much heat flows is captured by a 200-year-old rule, Fourier’s law. In words: the heat flowing through a surface equals the conductivity times the steepness of the temperature change across it.

The equation, term by term

$$q = -K \frac{\partial T}{\partial z}$$

q is the heat flux — how much heat crosses one square metre each second (watts per m^2). K is the conductivity. $\partial T/\partial z$ is the temperature gradient: how fast temperature T changes as you move in depth z (it is just “rise over run” for the temperature-vs-depth line). The minus sign encodes “heat flows from hot to cold” — toward lower temperature.

In plain words

Think of a crowded room emptying through a door. The flow of people (q) is bigger if the door is wide (large K) and if the crowd is much denser inside than outside (steep gradient). Heat behaves the same way.

2.3 From a single rule to the heat equation

Fourier’s law tells us the flow *through one surface*. To track the temperature *everywhere over time*, we apply one more idea: bookkeeping. For any thin slice of soil, the heat that builds up inside it equals the heat flowing in from one side minus the heat flowing out the other. Combine that bookkeeping with Fourier’s law and you get the **heat equation** — the master equation of this whole study.

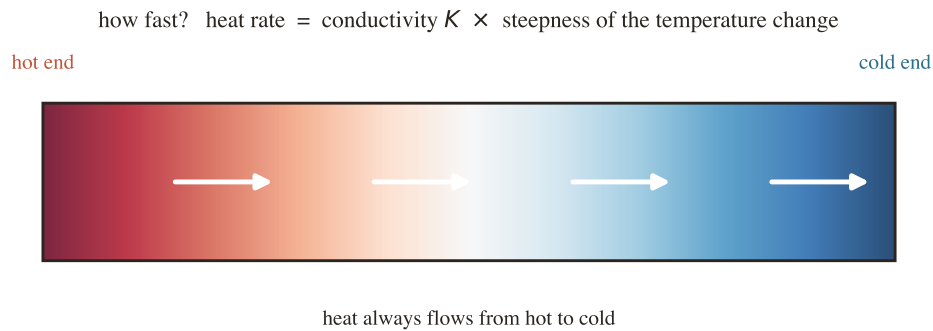


Figure 2.1. Conduction. Heat flows from the hot end to the cold end of a bar. How fast it flows depends on two things: how good a conductor the material is (K), and how steep the temperature change is along the bar. A steeper change, or a better conductor, means more heat per second.

The equation, term by term

$$\underbrace{\rho c_p \frac{\partial T}{\partial t}}_{\text{heat piling up in the slice}} = \underbrace{\frac{\partial}{\partial z} \left(K \frac{\partial T}{\partial z} \right)}_{(\text{heat in}) - (\text{heat out})}$$

Left side: how fast the slice's temperature rises ($\partial T / \partial t$), times how much heat that takes — ρ is density (how much stuff is packed in) and c_p is specific heat (how much energy to warm it by one degree). **Right side:** the net heat arriving by conduction — Fourier's law applied on both faces of the slice. If more heat comes in than leaves, the slice warms; if balanced, it holds steady.

Key takeaway

The heat equation just says: *temperature changes because more heat flows into a place than out of it*. Everything the computer does is solve this one equation, over and over, for the lunar soil column.

Chapter 3

The lunar setting: skin, depth, and the steady zone

3.1 Two sources of heat

The soil column is heated from both ends (Figure 3.1a). From *above*, sunlight pours in during the lunar day, and at night the surface cools by glowing away infrared light. From *below*, a tiny steady warmth rises from the Moon’s interior — the **basal heat flux**, written Q_b .

3.2 The skin depth: why the daily swing vanishes

The day–night temperature wave does not penetrate far. Each layer of soil absorbs and re-releases heat, smearing the wave out, so it shrinks fast with depth (Figure 3.1b). The depth over which it fades to insignificance is the **skin depth** — only a few centimetres on the Moon. By 80 cm down, the daily swing is smaller than our thermometers can detect.

In plain words

Press a warm hand on a thick quilt. The surface fibres warm, but a finger-width in, nothing changes. The Moon’s daily heating is that hand; it only reaches the top few centimetres. Everything deeper just feels the steady average.

3.3 The steady deep zone — where the measurement lives

Below the skin depth the soil is in a **steady state**: its temperature no longer wobbles with day and night. It only rises slowly and steadily with depth, because Q_b is forever trickling upward. In that steady zone the full heat equation collapses to the simple balance we will use to extract K_d .

The equation, term by term

$$\frac{\Delta T}{\Delta z} = \frac{Q_b}{K_d}$$

In the steady deep zone the temperature climbs at a constant rate with depth. That rate ($\Delta T/\Delta z$, the gradient) equals the upward heat flux Q_b divided by the conductivity K_d . Push fixed heat through a poor conductor (small K_d) and the temperature must climb steeply; through a good conductor (large K_d) it climbs gently.

Key takeaway

We can *read the gradient* off the buried thermometers, and we *know* Q_b from the mission. Rearranging, $K_d = Q_b/(\Delta T/\Delta z)$. That is the entire study in one line — everything else is doing this carefully and honestly.

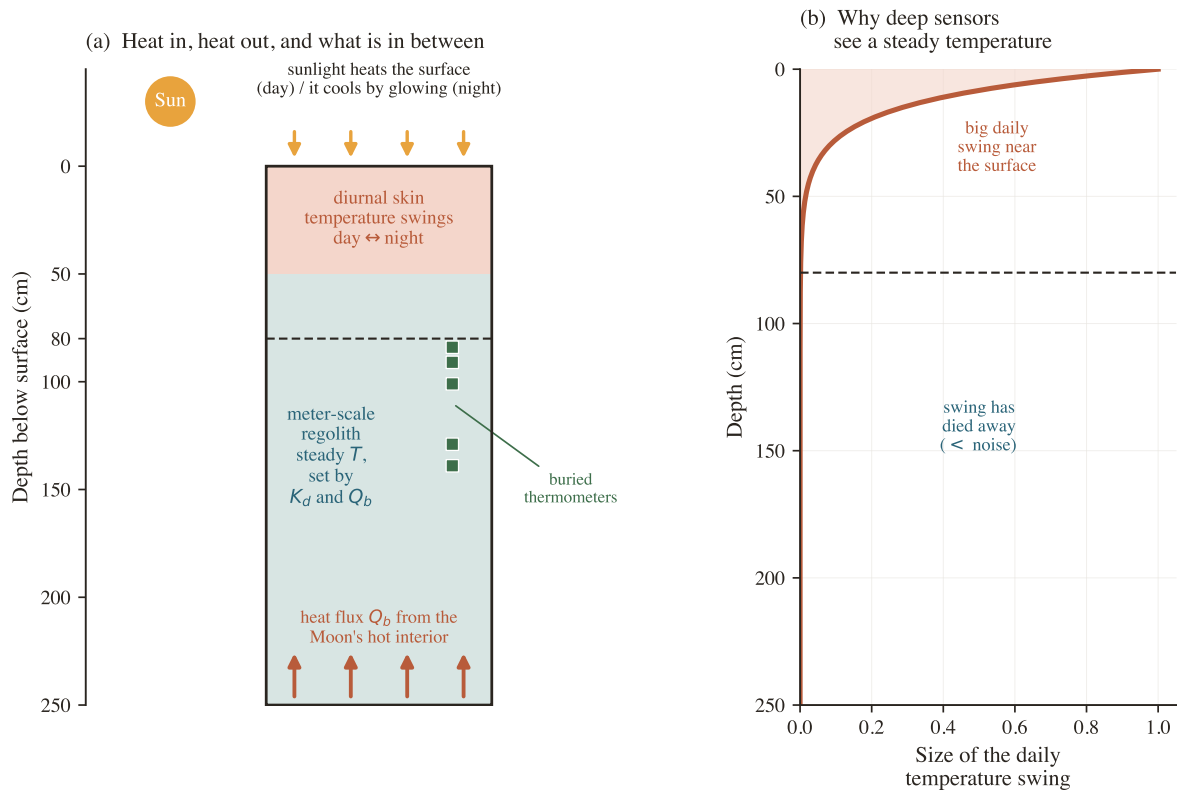


Figure 3.1. The lunar soil column. (a) Sunlight heats from above; the basal flux Q_b rises from below. The top zone (coral) feels the day–night swing; the deep zone (teal) is steady. Squares mark the buried thermometers. (b) The size of the daily swing collapses with depth — by 80 cm it is below the noise, which is why deep sensors read a steady value.

Part II

The Measurement

Chapter 4

The Apollo data: two boreholes, fifty years old

4.1 What was buried

On Apollo 15 (1971) and Apollo 17 (1972), astronauts drilled holes and lowered fibreglass tubes carrying thermometers to more than two metres depth (Figure 4.1). These “Heat-Flow Experiment” (HFE) probes recorded temperatures for several years. Some of the record was later lost, then painstakingly *restored from the original 1970s data tapes* in 2018. That restored record is our raw material.

4.2 Two honest housekeeping steps

Throw away the shallow sensors. The fibreglass stem itself conducts a little surface heat downward, contaminating the shallowest thermometers. Rather than guess a correction, we discard *every* sensor above 80 cm before looking at any answer. This is the **borestem cut**, and being strict about it up front is part of what makes the result trustworthy.

Use a settled temperature for each sensor. Each record has early disturbances (from drilling, equipment glitches, slow drift). For every thermometer we automatically find the long, flat, settled stretch near the end and average it — one reliable “equilibrium” temperature per sensor.

Key takeaway

After housekeeping, the entire dataset is: one trustworthy temperature and its depth, for each deep sensor. Apollo 15 gives 7 such points; Apollo 17 gives 16. Twenty-three numbers. The whole study is about extracting the most defensible conclusion from those twenty-three numbers.

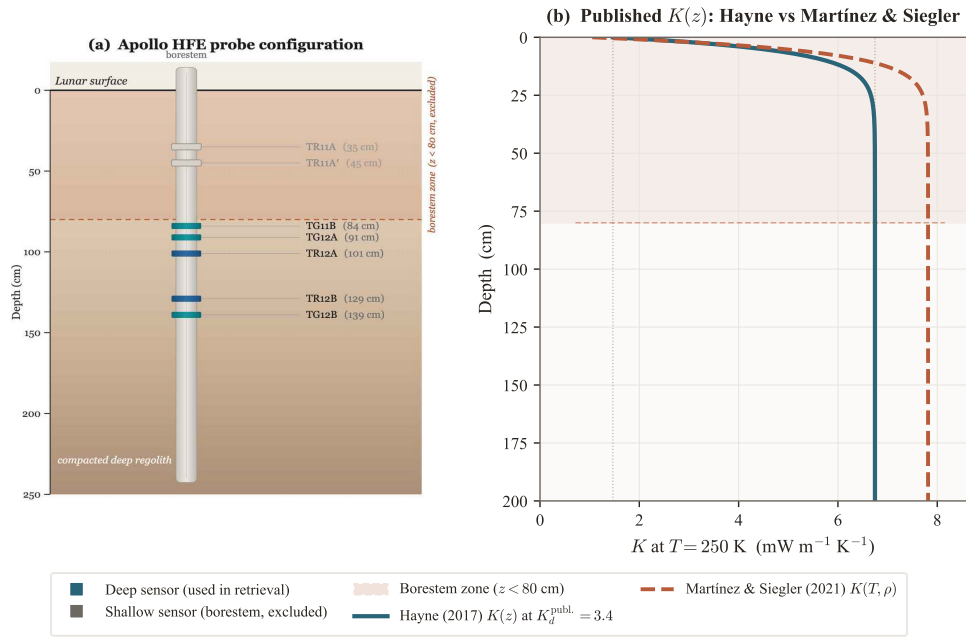


Figure 4.1. The probe. A fibreglass stem holding thermometers at several depths. The shaded band ($z < 80$ cm) is discarded (next section). The right panel shows the two textbook conductivity models we later compare against.

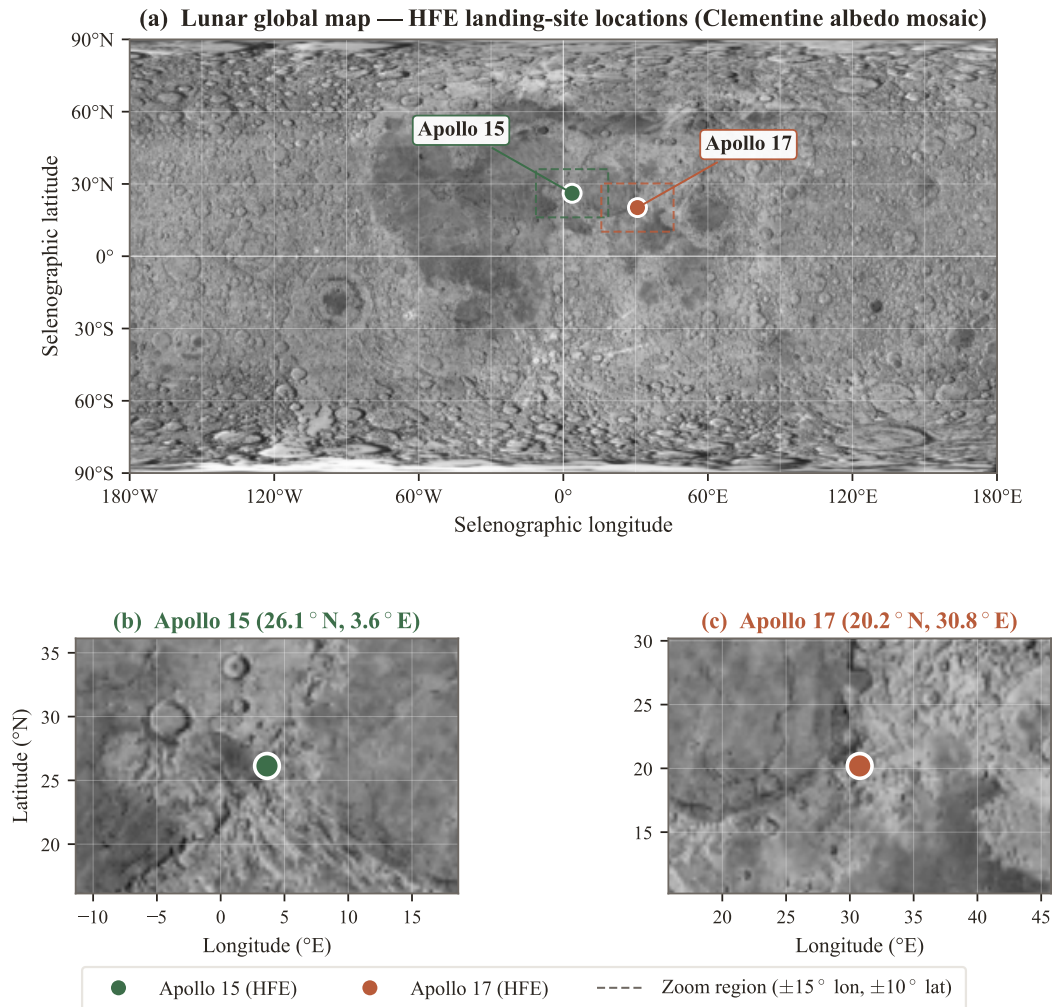


Figure 4.2. Where the boreholes are. Apollo 15 sits at a boundary between dark lava plains and bright highlands; Apollo 17 sits in a lava-filled valley. The different geology is one physical reason the two sites could have genuinely different conductivities.

Chapter 5

The forward model: using the equation on a computer

5.1 Why we need a computer at all

The simple gradient law ($\Delta T/\Delta z = Q_b/K_d$) holds only in the perfectly steady deep zone. To connect it to reality — including the day–night swing, the temperature-dependent soil properties, and the surface energy balance — we must solve the full heat equation. That cannot be done with pencil and paper, so we solve it numerically.

5.2 Chopping the world into pieces

A computer cannot handle a smooth, continuous column. So we **discretise**: chop the depth into many thin layers and time into many small steps (Figure 5.1). Within each layer the temperature is one number; each time step, we update every layer using the heat that flowed between it and its neighbours. Do this for enough steps and the smooth solution emerges.

In plain words

It is like animating a movie. Reality is continuous motion, but the film is a stack of still frames shown quickly. If the frames (our layers and time steps) are fine enough, the eye — and the physics — sees smooth motion.

A detail that matters: the grid is not uniform. The action is all near the surface (the skin), so we make the top layers very thin (2 mm) and let them grow thicker with depth. This puts the computing effort where the temperature changes fastest.

5.3 The two ends: boundary conditions

Solving the equation needs a rule at each end of the column.

- **Top (the surface):** at every instant, sunlight absorbed = infrared glowed away + heat conducted downward. This balance sets the surface temperature. It is not a fixed number — it changes through the lunar day.
- **Bottom (deep down):** heat flows up at the fixed rate Q_b . The column is fed from below, not sealed.

5.4 Settling down: equilibrium

Because the deep soil is such a good insulator, it takes a very long simulated time to “forget” wherever you started it and settle into its true repeating state. Getting this settling right turns

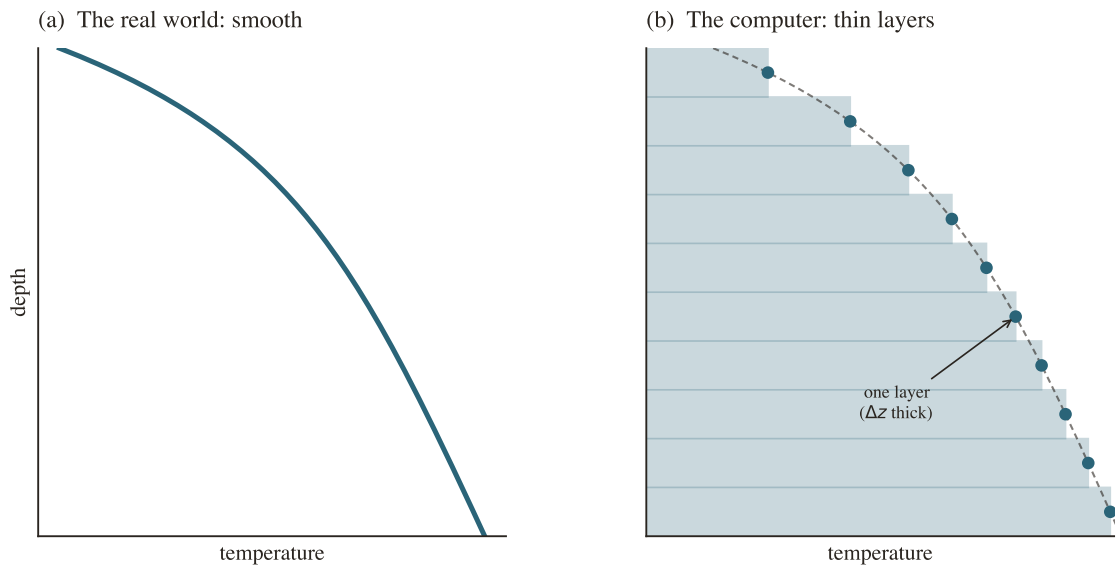


Figure 5.1. From continuous to computable. (a) The real temperature varies smoothly with depth. (b) The computer represents it as a stack of thin layers, each with a single temperature, updated step by step. Thin enough layers reproduce the smooth curve (dashed).

out to be the single most important technical point in the study — important enough that it has its own chapter (Chapter 10).

Part III

From Data to a Number (and its Uncertainty)

Chapter 6

The retrieval: turning temperatures into K_d

6.1 Guess, simulate, compare

We cannot measure conductivity directly, so we reverse the problem (Figure 6.1a): guess a value of K_d , simulate what temperatures it would produce, and see how well they match the real sensors. Try many guesses; keep the one that matches best.

6.2 Measuring the mismatch: RMSE

We need one number for “how badly does this prediction miss the data.” That is the **root-mean-square error**.

The equation, term by term

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (T_i^{\text{model}} - T_i^{\text{data}})^2}$$

For each of the N sensors, take the gap between model and measurement, **square** it (so misses in either direction count, and big misses count a lot), **average** over all sensors (the $\frac{1}{N} \sum$), then take the **square root** to return to units of degrees. Zero means perfect; bigger means worse.

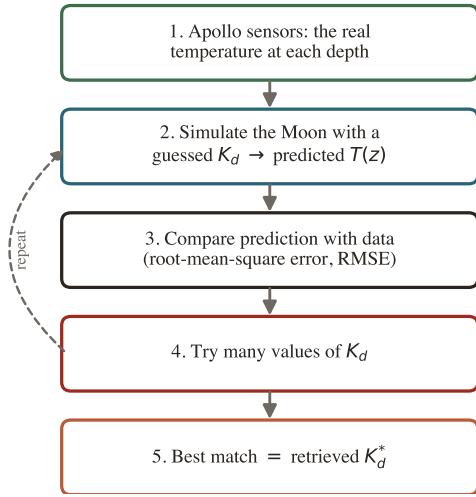
6.3 Reading the bowl

Plotting RMSE against the guessed K_d gives a U-shaped bowl (Figure 6.1b). Too-low K_d predicts too-steep a gradient (bad); too-high predicts too-gentle (also bad); the bottom is the best fit, written K_d^* . A clear bowl with a distinct bottom means the data really do pin down a value.

Key takeaway

The retrieved conductivity is simply the bottom of the mismatch bowl. Apollo 15’s bowl bottoms near 4.6, Apollo 17’s near 8.1 — already a hint that the two sites are different, and both above the single global value of 3.4.

(a) How a conductivity number is extracted



(b) The best fit is the bottom of the bowl

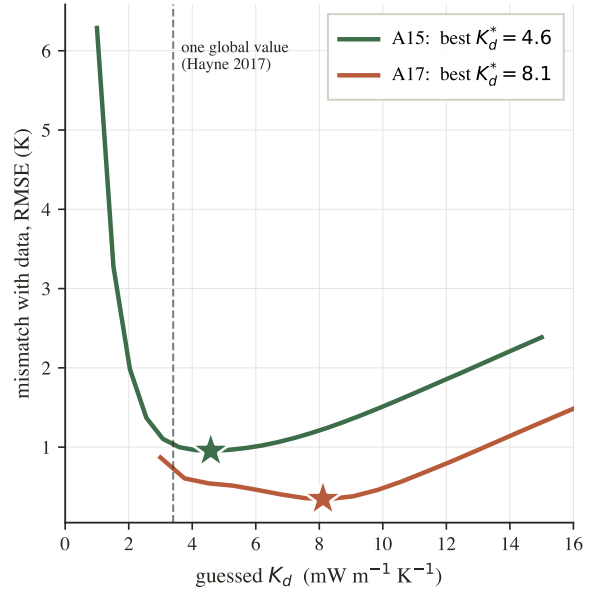


Figure 6.1. Extracting a number. (a) The guess–simulate–compare loop. (b) The real result: the mismatch (RMSE) traced against the guessed K_d forms a bowl; its lowest point is the best fit. Apollo 15 bottoms near 4.6, Apollo 17 near 8.1; the dashed line is the single “global” value (3.4) in use today.

Chapter 7

Uncertainty I: the bootstrap

7.1 The question

The bowl gives one best number, but with only 7 or 16 data points, how sure can we be? If we had drilled the holes a little differently, would we get a wildly different answer? We need an honest error bar.

7.2 The trick: resample your own data

We cannot redo the Apollo missions. But we can *pretend* to, using a clever statistical trick called the **bootstrap** (Figure 7.1). The idea: treat your real sensors as a bag of tickets, and draw a new “mission” from the bag *with replacement* — meaning after each draw you put the ticket back, so some sensors get picked twice and others not at all. Each such fake dataset is slightly different, so each gives a slightly different K_d . Do this 1,500 times and look at the spread of answers.

In plain words

Why does this work? The spread you get from shuffling your own sample mimics the spread you would get if you could repeat the whole experiment many times. It lets a small dataset report its own reliability, with no assumptions about bell curves or formulas.

7.3 What we report

From the 1,500 answers we quote the middle (the median) and the range that holds 95% of them (the **95% confidence interval**). We also do this for the *difference* between the two sites — and find that the difference is positive (Apollo 17 more conductive) in about 99% of the resamples. “Significant at the 95% level” simply means: the answer almost never lands on the wrong side of zero.

Key takeaway

Bootstrap = measure your uncertainty by reshuffling your own data many times and watching how much the answer jumps around. Small jumps = confident; big jumps = uncertain.

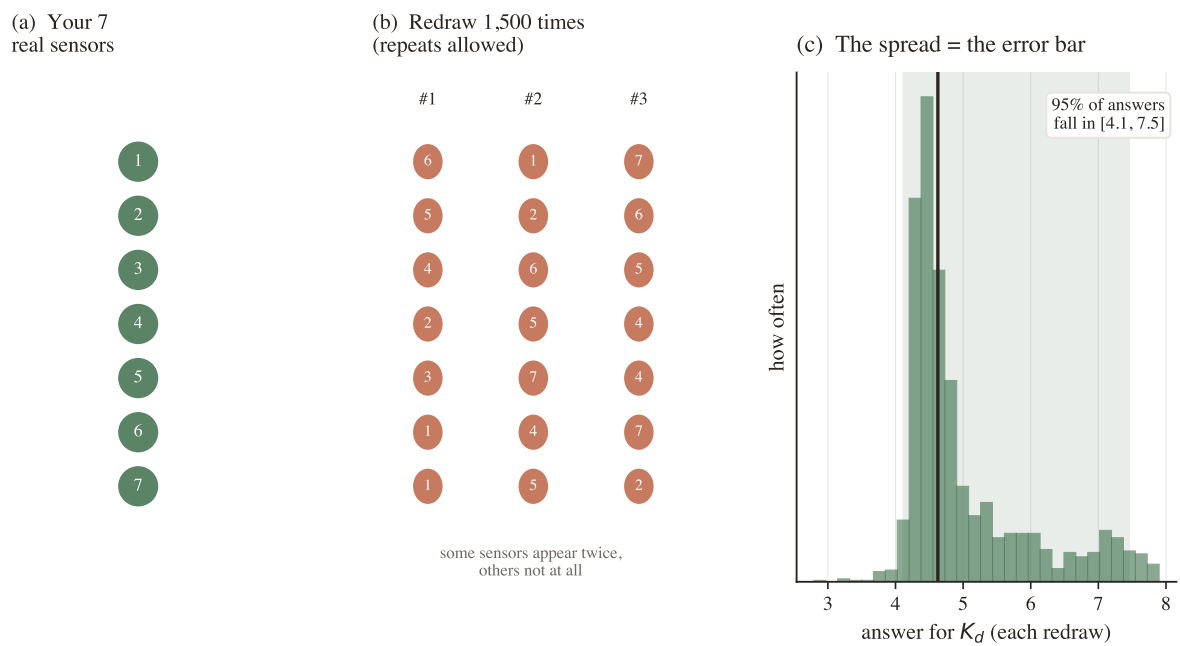


Figure 7.1. The bootstrap, from scratch. (a) Your real sensors. (b) Make many “resampled” datasets by drawing with replacement — repeats allowed, some left out. Each resample yields its own K_d . (c) The spread of those answers *is* the uncertainty: here, 95% of resampled answers for Apollo 15 fall between 4.1 and 7.5.

Chapter 8

Uncertainty II: Bayesian inference and MCMC

8.1 A second opinion, with a twist

The bootstrap makes no assumptions, which is its strength. But there is a second, complementary way to handle uncertainty that lets us fold in *what we already knew* about the basal heat flux Q_b . It is called **Bayesian inference**.

8.2 Prior, likelihood, posterior

Bayesian reasoning updates belief with evidence (Figure 8.1a):

- the **prior** is what we believed before — e.g. “ Q_b is probably around its published value, give or take.”
- the **likelihood** is what the new data say on their own.
- multiply them and you get the **posterior** — the updated belief, sharper than either alone.

The equation, term by term

$$\underbrace{P(\text{answer} \mid \text{data})}_{\text{posterior}} \propto \underbrace{P(\text{data} \mid \text{answer})}_{\text{likelihood}} \times \underbrace{P(\text{answer})}_{\text{prior}}$$

Read $P(A \mid B)$ as “how plausible A is, given B .” The posterior (updated belief about the answer, given the data) is the likelihood (how well each candidate answer explains the data) times the prior (what we believed beforehand). The “ $\propto$ ” means “proportional to” — we only need the shape, then normalise.

8.3 Why we need MCMC

For our problem the answer has two unknowns at once (K_d and Q_b), and the posterior is a 2-D landscape we cannot write down neatly. So we *explore* it with a computational method called **Markov Chain Monte Carlo (MCMC)**: imagine a hiker who wanders the landscape, preferring to step toward more-plausible spots but occasionally wandering downhill, leaving a trail of footprints. After enough steps, the density of footprints maps out the posterior. We do not need a formula — the walkers draw the picture for us.

8.4 The ridge: the deepest limitation of one borehole

The MCMC reveals something important (Figure 8.1b): the plausible answers form a long diagonal **ridge**, not a single peak. That is because a single borehole measures only the temperature gradient, which fixes the *ratio* Q_b/K_d — not K_d by itself. Halve both Q_b and K_d and the gradient

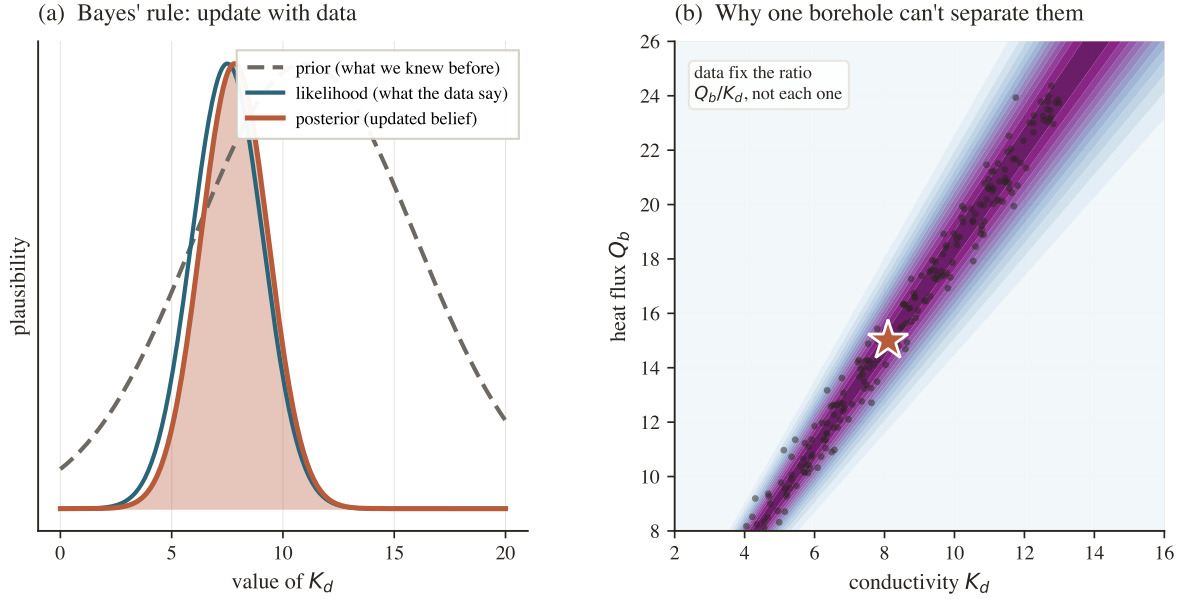


Figure 8.1. Bayesian updating and the ridge. (a) Prior (what we knew) times likelihood (what the data say) gives the posterior (updated belief), here sharper than either. (b) The 2-D landscape for our problem: plausibility is high all along a diagonal *ridge*. The data fix the *ratio* Q_b/K_d (the gradient), not K_d and Q_b separately. Each dot is an MCMC “footprint”; the star is the published value.

is unchanged, so the data cannot tell the difference. This “degeneracy” is why the absolute conductivity depends on what we assume for Q_b , and why the *difference between the two sites* is a sturdier result than either absolute value.

Key takeaway

Bayesian/MCMC = explore all answers consistent with the data *and* prior knowledge, by having computational “hikers” map the landscape. For us it confirms the bootstrap and exposes the K_d – Q_b ridge: one borehole pins the ratio, not the individual numbers.

Chapter 9

Choosing between explanations: the AICc

9.1 The danger of fitting too well

Suppose two stories explain the data: “the two sites have different conductivities” versus “they have the same conductivity but different heat fluxes.” How do we decide? We cannot just pick whichever fits the data more closely, because of a trap called **overfitting**: a model with more free knobs can *always* fit the dots better, even by chasing meaningless noise (Figure 9.1a).

9.2 Scoring fairly: misfit plus a penalty

The **Akaike Information Criterion** (the “c” is a small-sample correction, hence **AICc**) scores each model as its misfit *plus a penalty for every extra free parameter*. The best model is the one with the lowest score — it fits well *without* buying that fit through needless complexity.

The equation, term by term

$$\text{AICc} = \underbrace{N \ln\left(\frac{\text{misfit}}{N}\right)}_{\text{rewards a good fit}} + \underbrace{2k}_{\text{penalty per knob}} + \underbrace{\frac{2k(k+1)}{N-k-1}}_{\text{small-sample correction}}$$

N number of data points; k number of free parameters (knobs) in the model. The first term drops when the fit improves; the second and third rise as you add knobs. Lower total = better. Comparing models, only the *difference* in AICc matters: a gap of $\gtrsim 2$ is meaningful, $\gtrsim 10$ is decisive.

9.3 What it tells us

Scoring our three candidate models (Figure 9.1b), the “different K_d per site” model (M1) wins. The “same K_d ” alternatives score worse even though they were allowed to fit — the data prefer genuinely different conductivities, not a shared value propped up by adjusting other knobs.

Key takeaway

AICc = a fair scoreboard that rewards fitting the data but fines you for extra complexity. It picks the per-site-conductivity explanation over the one-size-fits-all alternatives.

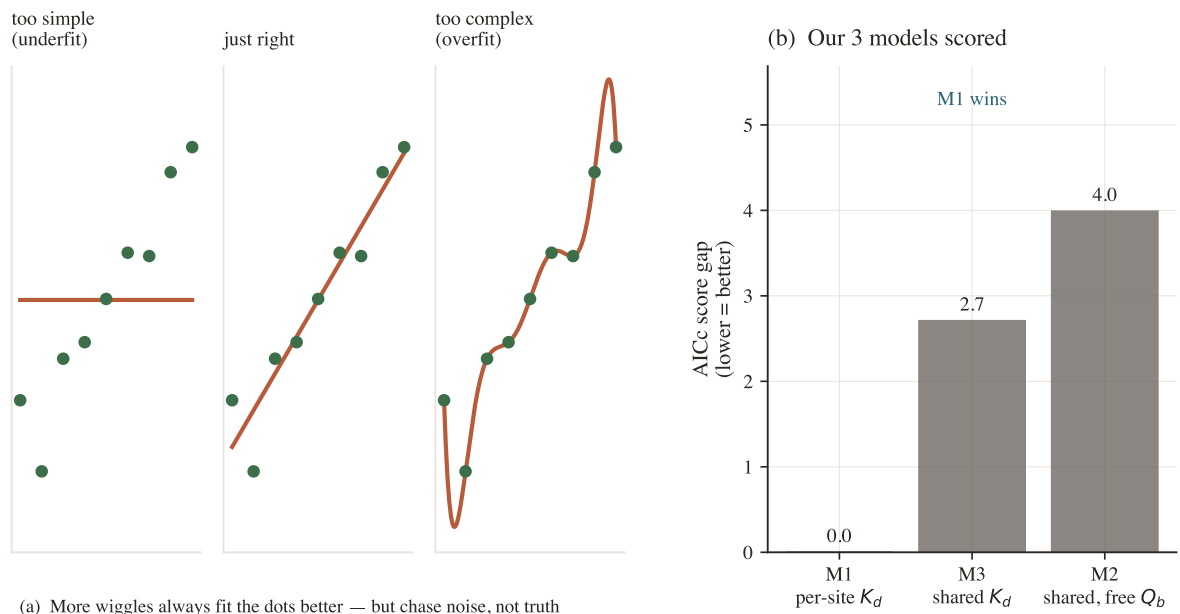


Figure 9.1. Overfitting and the AICc. (a) A too-simple model misses the trend; a too-complex one wiggles through every point, chasing noise; the middle is right. More parameters always lower the misfit, so misfit alone cannot choose. (b) The AICc adds a penalty for extra parameters and scores our three real models: M1 (different K_d per site) wins; M3 and M2 (shared K_d) score worse.

Part IV

The Result

Chapter 10

The mistake we found and fixed

10.1 The bug: an answer that depended on a guess

To start a simulation, the computer needs an initial deep temperature. The old code simply typed one in (250 K for Apollo 15, 255 K for Apollo 17) and ran for 30 lunar days to “settle.” But deep lunar soil is such a good insulator that true settling takes the equivalent of *roughly a thousand* lunar days. Thirty is far too few — so the deep temperature the model reported was mostly just the typed-in guess, and the conductivity read off from it was partly an echo of that guess rather than a fact about the Moon.

We proved this directly (Figure 10.1a): running the old method from three different starting guesses gave three different deep profiles, fanning apart by several degrees. An answer that moves when you change an arbitrary starting guess is not yet a measurement.

10.2 The fix: compute the settled state directly

Instead of “start from a guess and wait,” we compute the settled state directly, using the steady-state fact from Chapter 3: once settled, the *same* heat must flow through every layer, and that single condition fixes the entire deep temperature profile — no waiting needed. The method alternates a short simulation (to settle the fast surface skin) with a direct calculation (to fix the slow deep part), repeating until nothing changes.

In plain words

We did *not* simply “run longer.” Running a thousand lunar days for every one of thousands of trial values would take days of computer time. We used the steady-state shortcut to jump straight to the settled answer, at the cost of the old short run.

We then *verified* it: starting from wildly different guesses (240 vs 260 K) now gives the same answer to within two-hundredths of a degree, and a long check-run started from the settled state does not drift (Figure 10.1b). The answer is now a property of the data and the physics, not of a typed-in number.

10.3 What changed in the numbers

- **Apollo 15:** 4.86 $\rightarrow$ 4.58. Barely moved — its old guess (250 K) was close to the truth by luck.
- **Apollo 17:** 11.23 $\rightarrow$ 8.12. Dropped a lot — its old guess (255 K) was too warm and had inflated the conductivity.

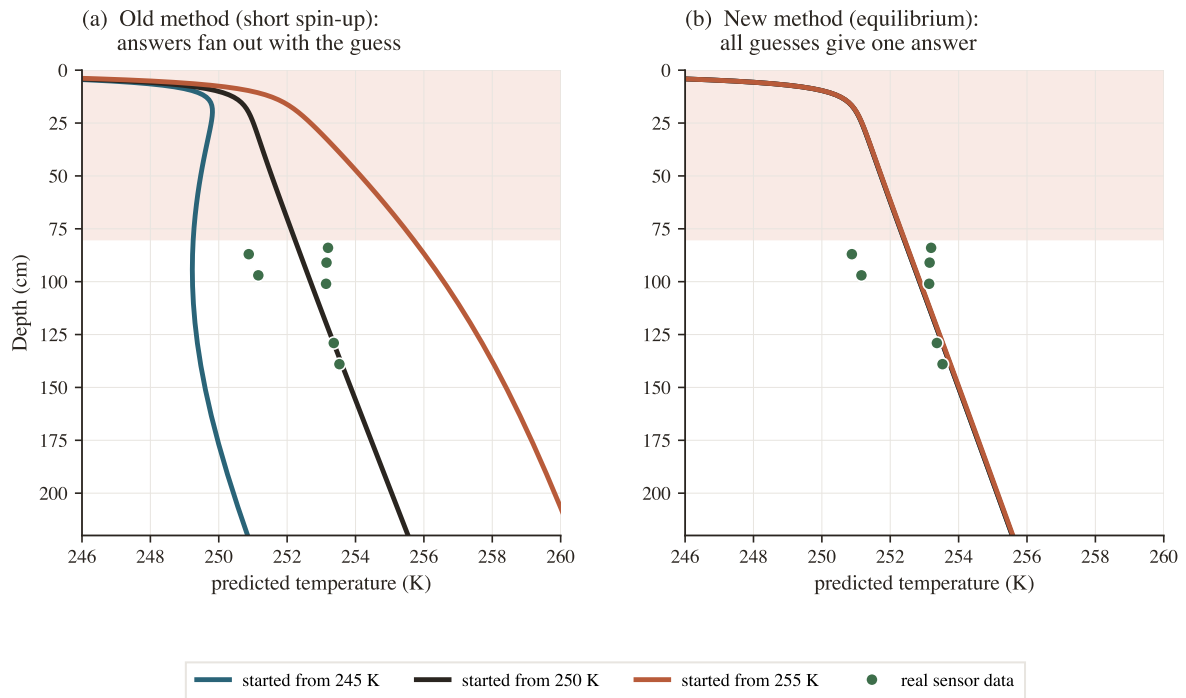


Figure 10.1. The bug and the fix (real model runs, Apollo 15). (a) Old method: starting from 245, 250, or 255 K, the predicted deep profile fans out — the arbitrary guess leaks into the answer. (b) New method: the same three starts land on a single curve through the real data. The result no longer depends on the guess.

Key takeaway

Fixing the bug made the science *stronger*: the difference between the sites became statistically significant, the Apollo 17 value moved into the range other evidence expects, and an independent model that used to contradict the Apollo 17 result now agrees with it. When removing a flaw makes independent checks line up, that is the signature of a correct fix.

Chapter 11

The results and what they mean

11.1 The headline numbers

	Apollo 15	Apollo 17
Retrieved K_d^* (mW m ⁻¹ K ⁻¹)	4.58	8.12
95% confidence interval	[4.1, 7.5]	[6.9, 9.0]
Difference (Apollo 17 – 15)	3.3, 95% CI [0.4, 4.6], significant	

For scale, these units (milliwatts per metre per degree): water is ~ 600 , styrofoam ~ 30 , and lunar regolith near 5–8 — an exceptional insulator. The fitted profiles through the real data are shown in Figure 11.1.

11.2 Why these values make physical sense

Apollo 15’s 4.6 sits between the global average (3.4) and orbital estimates (~ 7), matching what the soil’s fluffiness (porosity) predicts. Apollo 17 being ~ 1.8 times higher is explained by two effects pointing the same way: its soil is volcanic basalt (intrinsically more conductive than Apollo 15’s highland rock), and it is slightly more tightly packed. Neither effect needs to be extreme; together they cover the factor of 1.8 comfortably.

11.3 An independent sanity check

The retrieval used only *deep* temperatures. As a separate test, we checked whether the same models reproduce the *surface* temperature swing measured from orbit by the Diviner instrument (Figure 11.2). They do — confirming the whole simulation chain (sunlight, day length, surface balance) is set up correctly, independently of the deep retrieval.

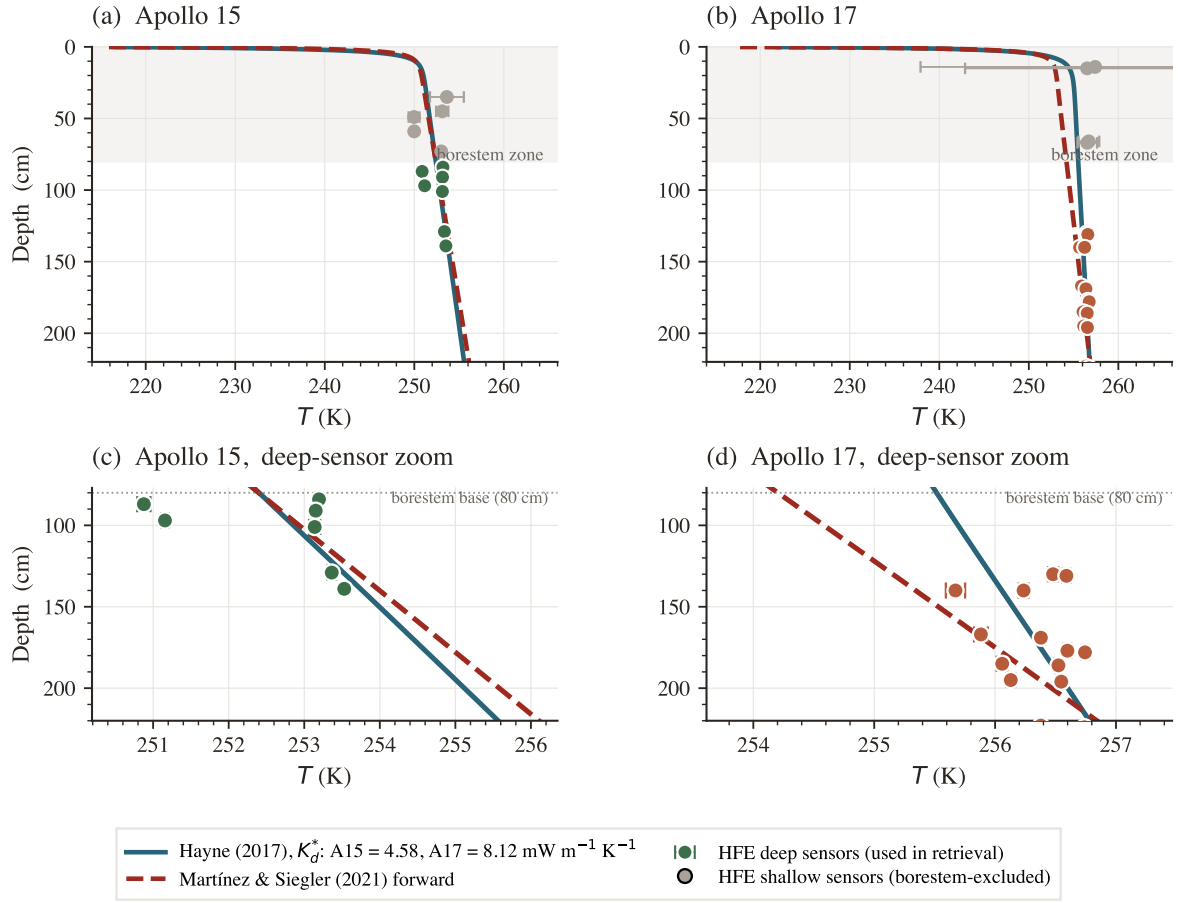


Figure 11.1. The fitted profiles. Dots are the real deep sensors; curves are the best-fit models. Bottom row zooms on the deep region used. Apollo 15 shows the *steeper* rise (lower conductivity needs a steeper gradient to pass its heat); Apollo 17, more conductive, rises more gently.

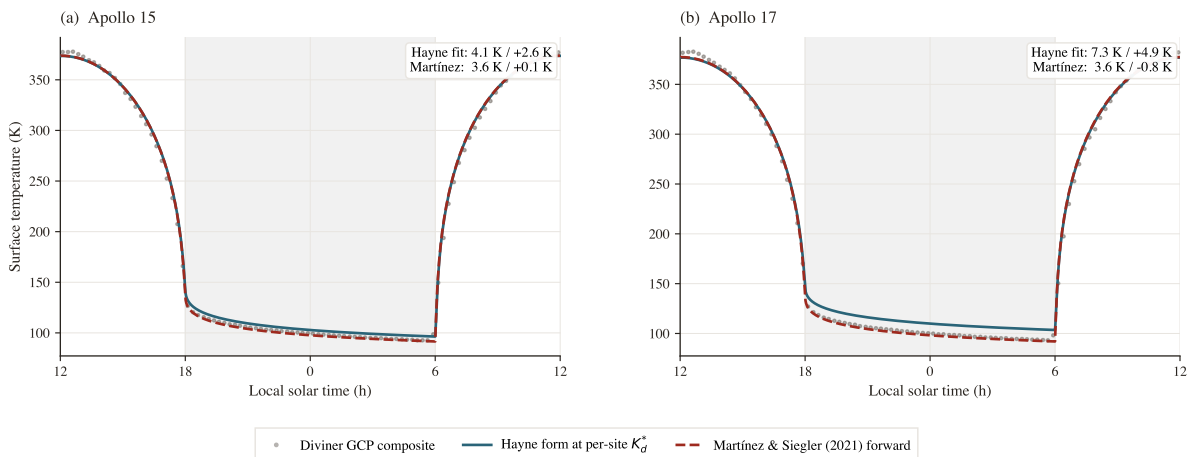


Figure 11.2. Independent check. Grey dots: surface temperature over a lunar day, measured from orbit. Curves: our models, which were tuned only to the deep sensors. The good agreement confirms the setup; this surface data was never used in the retrieval.

Chapter 12

Honest limits and the bottom line

A trustworthy study is clear about where it could be wrong. Ours has three main caveats:

1. **Absolute values lean on the assumed Q_b .** Because the data fix the ratio Q_b/K_d (the ridge of Chapter 8), revising the basal heat flux shifts K_d proportionally. The *difference* between sites is far more robust than either absolute number.
2. **Only two boreholes exist.** Two points cannot prove how conductivity varies across the whole Moon — only that these two places differ. The broader question needs new landed missions.
3. **Apollo 17's value is systematics-sensitive.** Its error bar is tight, but its accuracy depends on a few input choices (like the surface brightness coefficient), which we report openly in the paper's error budget.

Key takeaway

Rock-solid: the two sites need different conductivities, and Apollo 17 is the more conductive one. **Conditional:** the exact numbers, which depend on the assumed heat flux and the chosen model form. The paper is careful, throughout, to say which is which — and that honesty is what makes it credible.

Appendix A

How to run the code yourself

```
git clone https://github.com/rp3gregorio/Lunar-HFE.git
cd Lunar-HFE
python3 -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"
```

```
python scripts/pipeline/retrieve_kd.py          # the core retrieval
python scripts/figures/make_letter_figures.py  # the figures
python scripts/figures/make_results_figures.py
```

The pieces:

- **lunar/** — the engine. **solver.py** steps the heat equation; **equilibrium.py** is the settled-state solver of Chapter 10; **properties.py** holds the conductivity / density / heat formulas; **constants.py** stores every physical number with its source.
- **scripts/pipeline/** — the retrieval, bootstrap, Bayesian MCMC, AICc model comparison, and sensitivity checks.
- **scripts/figures/** — the figure generators, including **make_book_figures.py** and **make_primer_figures.py** which drew the teaching figures in this guidebook.
- **output/** — the committed results, so anyone can verify every number without rerunning.
- **docs/FLAG_REPORT.md** — the full audit trail behind Chapter 10.

Appendix B

Glossary of every term

Conduction	Heat spreading through solid material.
Thermal conductivity (K, K_d)	How easily heat passes through a material; K_d is the deep, settled value — our target. Small = good insulator.
Fourier's law	Heat flux = conductivity $\times$ temperature gradient.
Heat equation	The master equation: temperature changes where more heat flows in than out.
Density (ρ), specific heat (c_p)	How much material is present, and how much energy it takes to warm it.
Basal heat flux (Q_b)	The steady warmth rising from the Moon's interior.
Gradient ($\Delta T/\Delta z$)	Degrees of temperature change per metre of depth. Steep = good insulator (for fixed Q_b).
Diurnal skin / skin depth	The shallow zone where the day–night swing is felt; the skin depth is how far it reaches.
Steady state / equilibrium	The settled condition where temperature no longer changes from one lunar day to the next.
Boundary conditions	The rules at the column's ends: radiation balance on top, fixed flux Q_b at the bottom.
Discretisation	Splitting the smooth column into thin layers and time into small steps so a computer can solve the equation.
Borestem cut	Discarding sensors above 80 cm because the probe stem contaminates them.
Retrieval	Guessing K_d , simulating, and keeping the value that best matches the data.
RMSE	Root-mean-square error: one number for the typical model–data gap, in degrees. Smaller is better.
Bootstrap	Resampling your own data many times (with replacement) to measure how uncertain the answer is.
Bayesian inference	Updating belief by combining a prior with the data's likelihood to get a posterior.

Prior / likelihood / posterior

What you believed before / what the data say / the updated belief.

MCMC

Markov Chain Monte Carlo: exploring all plausible answers with random “walkers” that map out the posterior.

Degeneracy / ridge

When data fix only a combination of two unknowns (here Q_b/K_d), not each separately.

Overfitting

A too-complex model fitting noise instead of the true trend.

AICc

A model-comparison score: misfit plus a penalty for extra parameters. Lowest wins.

HFE

Heat-Flow Experiment — the Apollo instruments whose data we use.

A15 / A17

Apollo 15 and Apollo 17 sites.